

PRIMARY ISSUE CATEGORY: CEMENTING FAILURE

1. PROBLEM DEFINITION

A cementing failure covers losses of cement slurry into the formation during a primary cement job, and/or a poor cement bond (confirmed by CBL/VDL) that leaves inadequate zonal isolation, often producing an annular gas channel.

2. EARLY WARNING SIGNS

- Declining pump pressure without a corresponding rate change during the cement job (indicates losses).
- Reduced or absent returns at surface while pumping the tail slurry.
- A drilling-fluid loss history in the open hole immediately above the planned shoe depth.
- Post-job CBL/VDL amplitude attenuation below ~30-40% in the interval of interest.

3. ROOT CAUSES (FIELD_ALPHA HISTORY)

- Cement job pumped through a known loss-prone, high-permeability interval without loss-control additives.
- Slurry density/hydrostatic overbalance exceeding the formation's loss/fracture threshold.
- Washed-out, irregular hole geometry (confirmed by caliper log) reducing standoff and mud-cake quality.
- Inadequate mud conditioning / poor spacer design ahead of the slurry.

4. STEP-BY-STEP SOLUTION / RESPONSE PROCEDURE

Step 1: Continue the job safely — If losses occur mid-job, reduce pump rate rather than stopping outright, and complete the job with a top-up as planned to avoid a worse partial job.

Step 2: Run CBL/VDL — After waiting on cement (WOC), run a cement bond log and variable density log across the shoe interval to evaluate isolation quality.

Step 3: Interpret bond quality — Identify intervals with amplitude attenuation below ~30-40% as poor bond / likely channeling.

Step 4: Plan a remedial squeeze — For confirmed poor-bond intervals, plan a perforate-and-squeeze operation using a neat Class G slurry targeted at the specific depth.

Step 5: Verify the repair — Run a follow-up CBL/VDL and a casing pressure test to confirm the squeeze achieved adequate isolation before proceeding.

5. MATERIALS / PARAMETERS QUICK REFERENCE

CBL Amplitude Attenuation	Interpretation	Action
>70%	Good bond	No remedial action required
30-70%	Marginal bond	Evaluate against offset-well history; monitor
<30%	Poor bond / likely channel	Remedial squeeze cementing required

6. PREVENTION CHECKLIST FOR OFFSET WELLS

- Treat any drilling-fluid loss event within 500 m of a planned shoe as a trigger to add LCM additive to the cement slurry design.
- Reduce lead slurry density (e.g. 13.5-14.0 ppg) when setting a shoe within a known loss-prone interval.
- Cross-check LWD caliper data against the planned cement volume and adjust for excess washout volume.
- Make CBL/VDL logging mandatory (not contingent) for any shoe set in a known loss/washout interval.

7. FIELD CASE REFERENCES

Well	Depth	Outcome
W004	2600 m (shoe)	55 bbl slurry lost; poor bond 2540-2600 m; remedial squeeze at 2570 m restored isolation

This Issue Resolution Playbook is a synthetically generated reference document produced for the NWIS prototype, consolidating lessons across Field_Alpha and Field_Alpha-adjacent wells. Not associated with an actual drilling operation.